Pedro Henrique Muralha Paranhos

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EDUCATION

Federal University of Rio de Janeiro, UFRJ — Rio de Janeiro, RJ
Bachelor's degree in Mechanical Engineering — June 2023 - December 2027 (projected)

Federal University of Rio de Janeiro, UFRJ — Rio de Janeiro, RJ
Bachelor's degree in Electrical Engineering — January 2021 - June 2023 (Incomplete)

EXPERIENCE

Project Intern — Rio de Janeiro, RJ
Moove (Cosan Lubricants and Specialties SA) — September 2024 – December 2025

- Technical support for highly complex projects, including the optimization of grease and oil mixers, pipeline automation with pneumatic valves, and civil engineering infrastructure.
- Direct leadership in smaller-scale projects, from the quotation phase to installation, focusing on instrumentation and civil renovations.
- Implemented an official engineering project checklist, increasing the efficiency and quality of technical deliverables.
- Supported weekly safety presentations and project status updates.
- Assisted in developing a Capex plan.
- Managed the purchasing process and initiated hiring procedures to ensure the physical and financial schedule of deliveries is met.
- Provided direct support to designers in the development of technical drawings in AutoCAD, ensuring fidelity between the designed scope and the execution in the field.

PROJECTS

Complex AI Agents with Python
https://github.com/pedromparanhos-sys/complex-ai-agents-langgraph-course

- Built 5 AI agents in Python using LangGraph and LangChain (GPT-4o), implementing tool-calling (ReAct pattern), conversational memory, and human-in-the-loop workflows, orchestrated as state graphs with conditional routing.
- Built an end-to-end RAG (Retrieval-Augmented Generation) pipeline, integrating PyPDF, OpenAI embeddings, and ChromaDB as a vector store to answer document-grounded questions with source citations.

Statistical Pairs Trading with Kalman Filter (Python)
https://github.com/pedromparanhos-sys/pairs-trading

- I developed a complete pairs trading pipeline for S&P 500 stocks, implementing cointegration tests (Engle-Granger, Johansen) and a Kalman Filter for dynamic hedge ratios, recovering spread stationarity where static methods failed (ADF p-value 0.011 vs 0.63).
- I validated the strategy via walk-forward out-of-sample across 12 six-month windows (2018–2026), obtaining an average Sharpe ratio of 0.79 with 91.7% of windows showing positive results.
- I have honestly documented limitations (costs consuming 62% of the gross alpha) with a complete mathematical appendix of the derivations (cointegration, Kalman, Ornstein-Uhlenbeck process).

Predictive Microstructure Signal in Order Book (Python, Binance API)
https://github.com/pedromparanhos-sys/orderbook-signal

- I built a real-time collection pipeline via WebSocket of 3M+ snapshots of the L2 order book (BTC/USDT), with feature engineering for Order Book Imbalance and Trade Flow Imbalance.
- I identified monotonic signal decay between prediction horizons via purged k-fold cross-validation (AUC 0.89 at 1s, 0.61 at 30s), replicating results from the microstructure literature (Cont, Stoikov & Talreja, 2010).
- I quantified a cost/signal ratio of 17:1 in a backtest with real transaction costs, concluding that the statistical edge exists but is unfeasible with taker execution, a conclusion validated with sensitivity analysis across 6 thresholds.

Leveraged Buyout (LBO) Model - Fleury S.A. (B3: FLRY3)
https://github.com/pedromparanhos-sys/lbo-fleury

- I built a complete LBO model for Fleury S.A. (B3: FLRY3), integrating Excel and Python, with a projected return of 2.74x MOIC / 22.3% IRR on a capital structure of R$ 13.96 billion.
- I reconciled historical financial data directly from the CVM (Brazilian Securities and Exchange Commission), identifying leverage distortions between financial debt and lease liabilities (IFRS 16).
- I developed a Monte Carlo simulation engine in Python (10,000 scenarios, correlated variables), integrated with Excel via openpyxl, to quantify risk-return with 6 decimal place precision.

SKILLS

- **Software:** Microsoft Office, MS Project , Power BI, SolidWorks, Python (Anaconda, IPython, Jupyter Notebook, Pandas, NumPy, Matplotlib, SciPy, Seaborn, Scikit-Learn, Statsmodels, Polars, Plotly), LangChain, LangGraph, SQL, HTML, CSS, JS, Claude Code.
- **Languages:** Portuguese (native), English (advanced).